Strategies for solving full-rank, ill-conditioned Linear systems and Eigenvalue problems

MAT226B Final Project

By

Ashwin Amalaruban

Student ID: 922850545

Instructor: Dr.Thomas Strohmer

UNIVERSITY OF CALIFORNIA, DAVIS

DAVIS, CA-95616.

1. Solving Ax=b for well-posed but ill-conditioned matrices

Stokesian Dynamics gives a simple relationship between the velocities and the forces of particles as F = RU. R is a 11N x 11N matrix, while for the case of rigid bodies we consider 6N x 6N matrix where translational and rotational forces are important.

1.1 Brief overview of Stokesian Dynamics

When the translational and rotational velocities on particles are known, the corresponding forces and torques acting can be determined by using the Grand Resistance Matrix, R. Similarly, when the forces and torques acting on particles are known, translational and rotational velocities can be determined.

$$\begin{pmatrix} F^\alpha \\ L^\alpha \\ S^\alpha \end{pmatrix} = R \begin{pmatrix} U^\alpha - u^\infty \\ \Omega^\alpha - \Omega^\infty \\ -E^\infty \end{pmatrix}$$

U$-u^\infty$ $and\ \Omega - \Omega^\infty$ are vectors of length 3N for N particles. $-E^\infty$is a vector of length 9N. F and L are vectors of length 3N. S is the stresslet of length 9N[1].

Resistance matrix, $R = \begin{pmatrix} A & \tilde{B} & \tilde{G} \\ B & C & \tilde{H} \\ G & H & M \end{pmatrix}$ is of size 11N x 11N. For the case of rigid bodies we consider the first 6N x 6N matrix, $R_{FU} = \begin{pmatrix} A & \tilde{B} \\ B & C \end{pmatrix}$

Each of the component matrix contains further sub-matrices. The grand resistance matrix R contains nine sub-matrices, out of which six are said to be independent. The elements A, B, C and $\tilde{B}$ are rank 2 tensors, while elements $\tilde{G}$ and $\tilde{H}$, G and H are rank 3 tensors. The element M is a rank four tensor.

$$A_{ij} = A_{ji}, C_{ij} = C_{ji}, B_{ij} = \tilde{B}_{ji}, M_{ijkl} = M_{klij}, G_{ijk} = \tilde{G}_{kij}, H_{ijk} = \tilde{H}_{kij}$$

Here, the far-field resistance matrix, R^{ff} is dense, depends on the configuration of particles and is obtained for all particle pairs using Faxen's laws. The near-field matrix, R^{nf} is sparse and accounts for all pairs within a critical radius r* from a particle centre.

1.2 Aggregates and increasing condition number

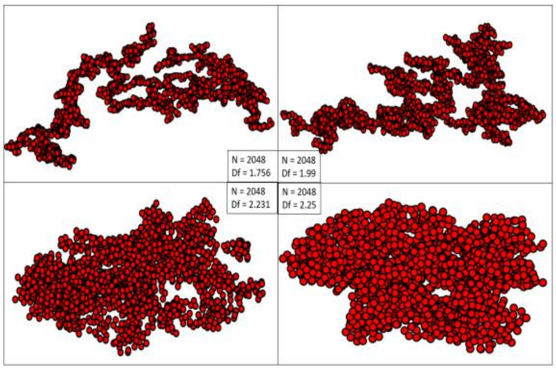

Figure 1: Varying fractal dimensions for N=2048 spheres in an aggregate

The aggregates are characterized using fractal dimension, D_f follows the scaling relationship, $\mathrm{N} = k_f \left(\frac{R_g}{a}\right)^{D_f}$ where a is the primary particle radius and k_f is the fractal pre-factor. Fractal dimension of 2.49 is denser, while the one with 1.76 is sparse and have elongated branches. *Figure 2* shows that the condition number increases with both N and fractal dimension.

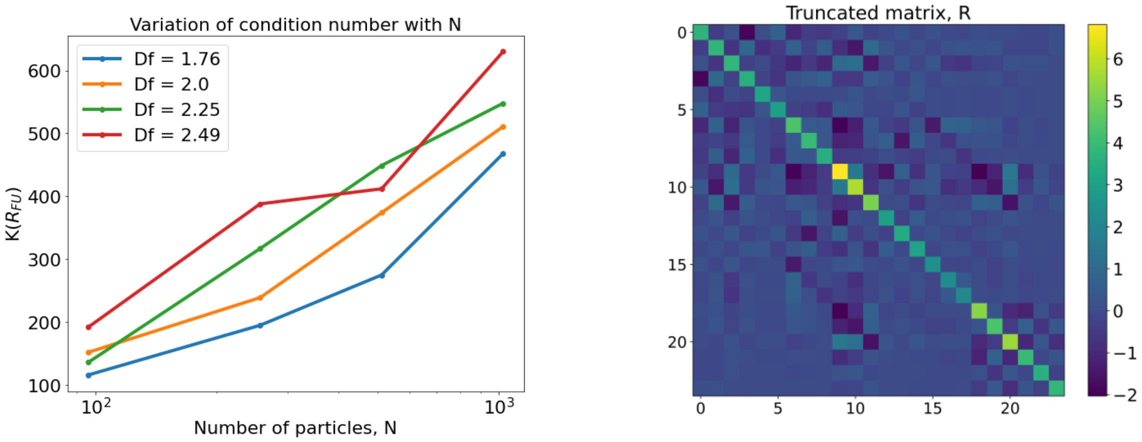

Figure 2: a) Variation of condition number of the resistance matrix, R_{FU}(6N x 6N) with the number of particles in an aggregate, N. b) Truncated matrix, $R \in \mathbb{R}^{24 \times 24}$ of $R_{FU} \in \mathbb{R}^{576 \times 576}$

1.3 Application using the concepts from MAT226B for Ax=b

Major objectives of this exercise were to reduce the time-complexity and computational resources required for higher condition number matrices. Generating the resistance matrix itself is time consuming $O(N^3)$, whereas solving RU=F took 20% of the total time for the entire process. Various iterative solvers such as Conjugate gradient, GMRES, Steepest Descent, LSQR and Craigs' along with SVD, Tikhonov were used.

Problem: A unit force is given in z-direction such that F=$[001001001\ ...]^T$ and the sphere coordinates are known. Velocities need to be determined to update the particle position every timestep. One of the major advantages that could emerge with applying a few of these methods is that we could use the solution from previous timestep for faster convergence.

1.4 Results

1.4.1 Conjugate Gradient and its variants

Objective: Comparison of simple Conjugate Gradient, Preconditioned Conjugate Gradient and CG with restart for a tolerance of 10^{-4} in residual norm.

Preconditioners were obtained using Landweber-Richardson iteration (extreme eigen values), Jacobi iteration, Gauss-Seidel iteration and incomplete Cholesky factorization. Gauss-Seidel iterations yielded a preconditioner, M which resulted in a convergence rate decrease before saturating at $\approx 10^0$ and not shown in the figure below.

Richardson iteration hardly reduced the condition number, while Jacobi iteration reduced the condition number of $M^{-1}A$ to an extent. Preconditioner using Jacobi iteration helped conjugate gradient converge faster as a reason.

Incomplete Cholesky factorization got the best preconditioner which ensured a $K(M^{-1}A) \approx 1$ and converged within 2 iterations. Comparison of conjugate gradient along with preconditioned CG is provided in *Figure 3*.

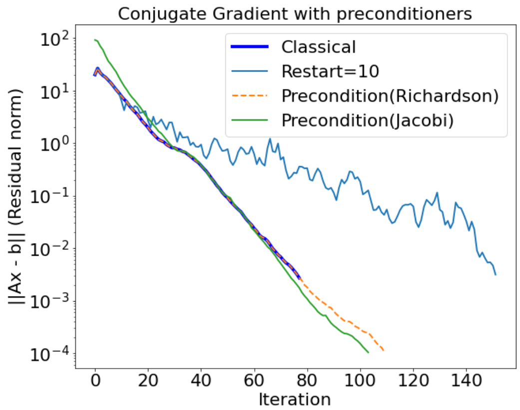

Figure 3: Convergence rate of conjugate gradient and its variants for $R_{FU} \in \mathbb{R}^{3072\times3072}$

Conjugate Gradient with restart=10 worsened the convergence after 10 iterations and was not useful.

1.4.2 GMRES

Objective: Convergence rate with changes in restart for $R_{FU}(6N \times 6N)$ and $R(11N \times 11N)$

At each iteration, Lanzcos process was used to orthonormalize the subspace. After a fixed number of steps (restart parameter of 5,10,25,50 and 100), the algorithm was restarted.

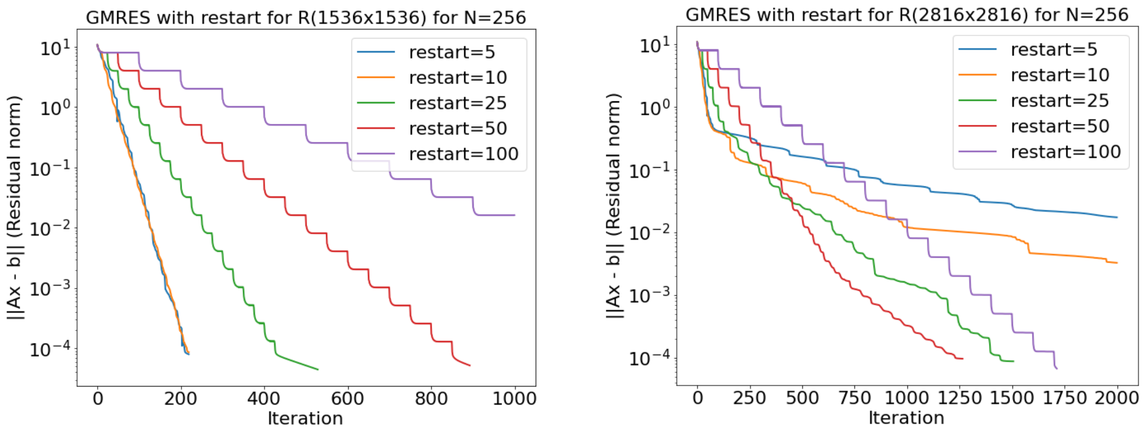

Figure 4: Convergence rate of GMRES for two different cases a) 6Nx6N resistance matrix for N=256 with restarts of 5,10,25,50,100 iterations, b) 11Nx11N resistance matrix for N=256 with restarts of 5,10,25,50,100 iterations

For a resistance matrix with $K_{R_{FU}}(A) = 388$, the smaller the restart parameter faster the convergence. However, for cases when the entire resistance matrix was considered such as $K_R(A) = 1212$ in *Figure 4*, smaller restarts converged fast initially before stagnating, although restart=50 initially with a slow convergence got better results.

1.4.3 Comparison of various solvers

Objective: i) Time complexity and efficiency of solvers

ii) Compare convergence rate of iterative solvers

Process time for the various solvers were compared to the numpy.linalg solver used previously with the Stokesian Dynamics algorithm. As N increases, Conjugate gradient and GMRES were far superior in reducing time whereas Steepest Descent can be adopted when computational resource become restricted. It is important to note that, GMRES is being used in recent versions of Stokesian Dynamics called "Accelerated Stokesian Dynamics", where Fast Fourier Transformation is used in the generation of the Resistance matrix[2]. *Figure 5b* shows that Steepest Descent isn't good for accuracy but can be used when N is very large as there its requires less memory.

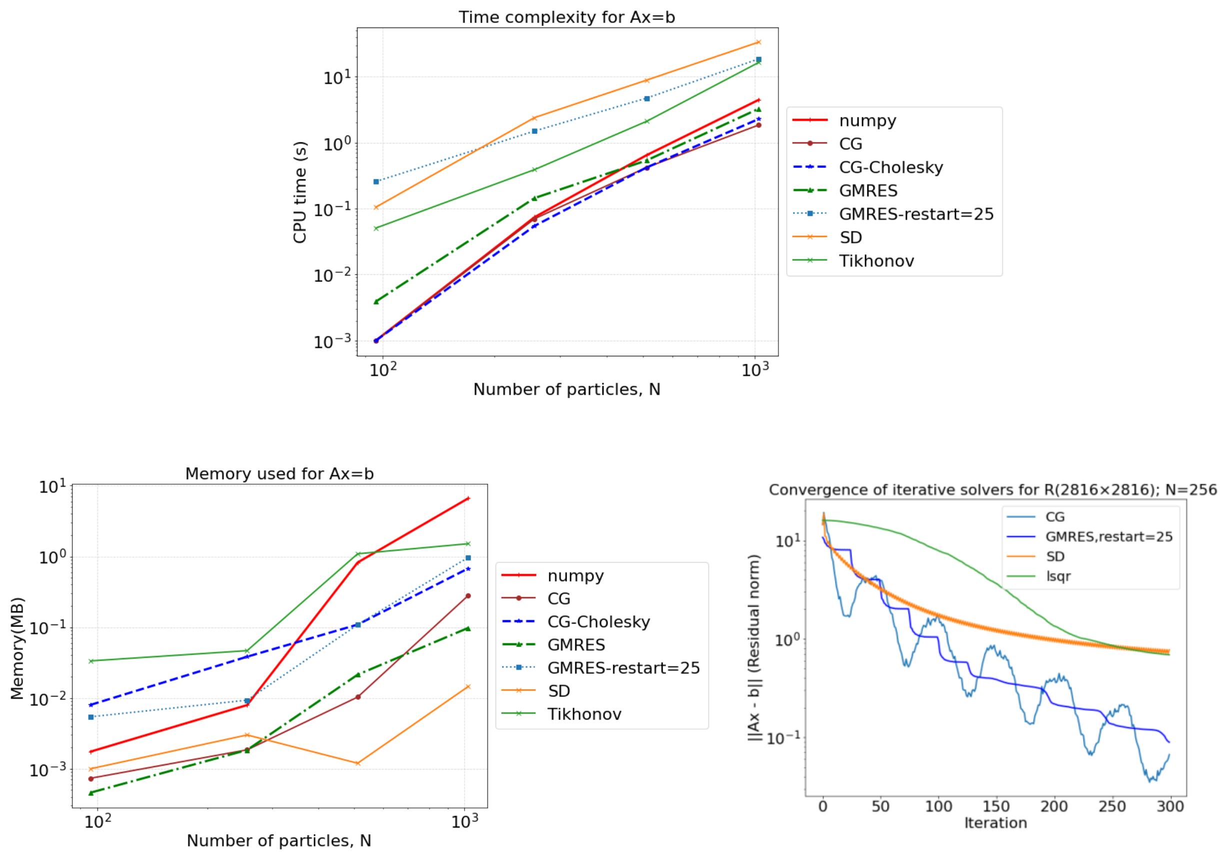

Figure 5: a) Time complexity and b) memory usage for various solvers of Ax=b (6Nx6N); c) Convergence rate of various iterative solvers

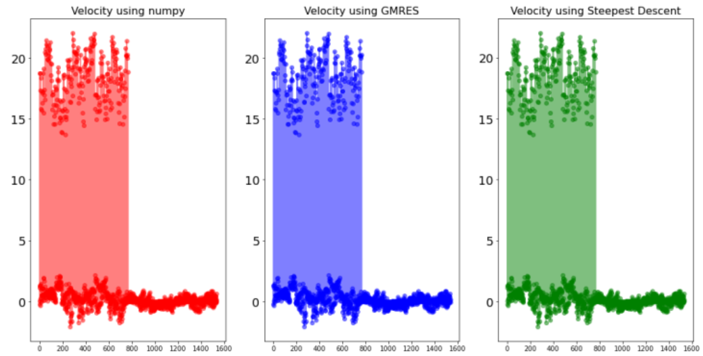

Figure 6: Velocity solution using numpy, GMRES and Steepest Descent.

2 Eigenvalue problem

Eigenvalues of the resistance matrix can be used to obtain preconditioners for Conjugate gradient method. Other than this, to account for the Brownian forces in Stokesian dynamics, a Chebyshev polynomial approximation is used to compute the square root of resistance matrix which requires the knowledge of the spectrum[3]. QR algorithm, Lanczos algorithm and SVD are used, and their performance is compared with respect to accuracy and efficiency.

2.1 Overview

i) Simple QR algorithm for matrices of different condition number and different size

ii) QR algorithm with shifts (Rayleigh and Wilkinson) and applying QR after Lanczos.

iii) Extreme eigenvalues using Lanczos, Lanczos with QR and Lanczos with SVD.

iv) Time complexity of all these methods for the entire spectrum.

2.2 Simple QR algorithm for matrices of different condition number and different size

Resistance matrices of the same size, but different condition number are compared using the classical QR algorithm. Convergence here is obtained using norm of the difference in eigenvalues between iterations. *Figure 7* suggests that convergence improved as the condition number increased.

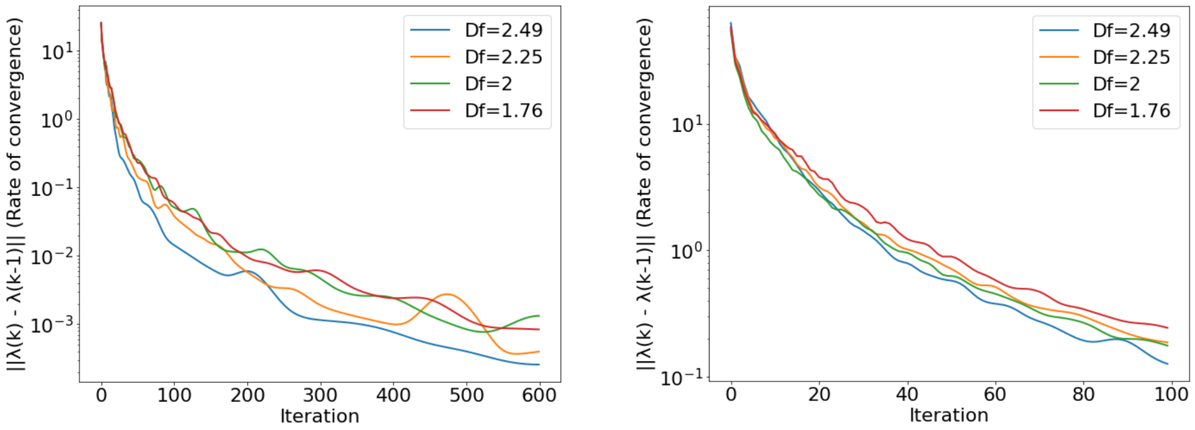

Figure 7: Convergence rate(Norm of difference in eigenvalues) of QR algorithm for the case of a) N=96; $A \in \mathbb{R}^{576\times576}$(tolerance=$10^{-5}$ and maximum iterations=600) and b) N=512; $A \in \mathbb{R}^{3072\times3072}$ for four different fractal dimensions(different condition numbers)(tolerance=10^{-5} and maximum iterations=100)

2.3 QR algorithm with shifts (Rayleigh and Wilkinson) and Lanczos with QR

The shift parameter for Rayleigh shift, $\mu^{(k)}$ is obtained every iteration from the bottom-right element, $A^k(n,n)$ where n x n is the size of the matrix. The shift parameter for Wilkinson shift, $\mu^{(k)}$ is the eigenvalue of the bottom-right 2x2 submatrix, A^k.

Lanczos algorithm is used to obtain a tridiagonal matrix and QR algorithm reduces it to a diagonal form which gives the eigenvalues.

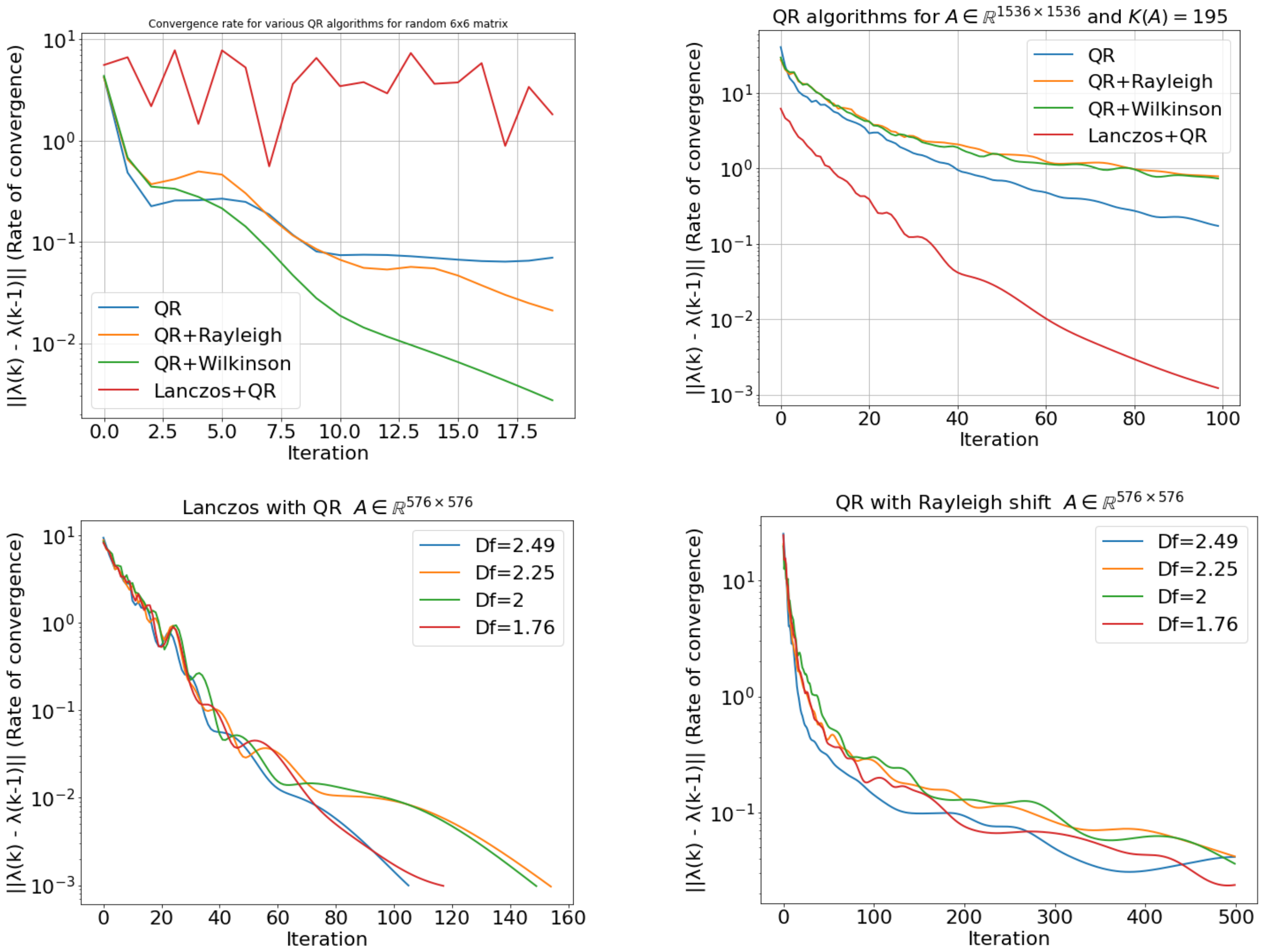

Figure 8: Convergence rate of a) various QR algorithms for a random 6 x 6 matrix(tolerance $=10^{-3}$ *,maximum iterations=20 and k=6 for Lanczos+QR), b) various QR algorithms for N=256 and* $D_f = 1.76$*(tolerance* $=10^{-3}$*,maximum iterations=100 and k=96 for Lanczos+QR) c) Lanczos with QR for different* D_f *and N=96(tolerance* $=10^{-3}$*,maximum iterations=100 and k=24), c) QR with Rayleigh shift for different* D_f *and N=96(tolerance* $=10^{-3}$*;maximum iterations=100)*

Like *section 2.2*, convergence rate here compares eigenvalues between iterations. A sample 6 x 6 matrix ensured rate of convergence improvised with shifts and better accuracy with Wilkinson shift in *Figure 8*. QR algorithm with shifts has not worked well with the resistance matrix of N=256 compared to the classical method. However, Lanczos along with

QR has a better accuracy compared to other algorithms or variations of QR as the dense form is condensed to a tridiagonal matrix.

Bottom images of *Figure 8* compares the rate of convergence for the matrix of same size and different condition numbers using Lanczos + QR and QR with Rayleigh shift. Eventhough, matrix of relatively high condition number converges faster, there is no correlation between the condition number and convergence.

2.4 Extreme eigenvalues using Lanczos, Lanczos with QR and Lanczos with SVD

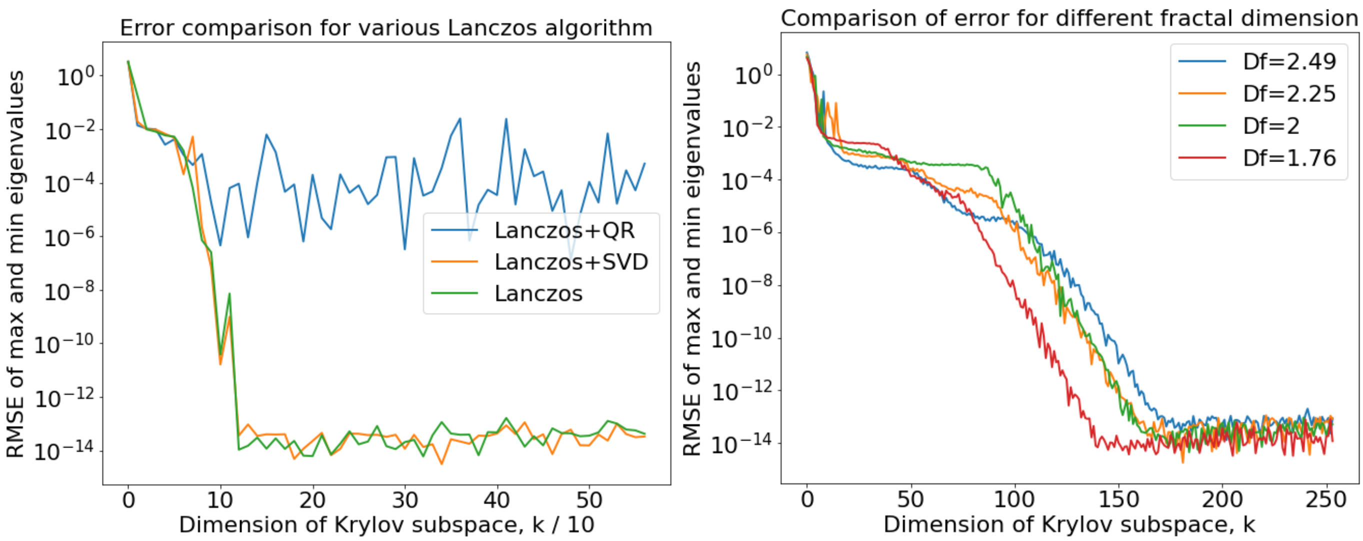

Figure 9: a) RMSE comparison for the various Lanczos algorithms by varying the dimension of subspace(maximum iterations=100 for the QR algorithm after Lanczos) for $A \in \mathbb{R}^{576\times576}$,b) RMSE comparison of Lanczos by varying the dimension of subspace for $A \in \mathbb{R}^{1536\times1536}$

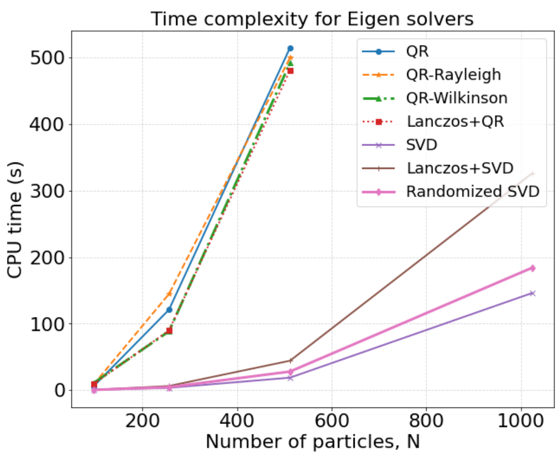

Figure 10: Time complexity of various eigensolvers

3 Conclusion

Conjugate gradient, GMRES or Preconditioned CG have been helpful in reducing the time complexity of the resistance problem, while Steepest Descent can also be used when N is large with reduced accuracy. Eigenvalues of the resistance matrix were sought to understand the better algorithms rather than using them for this application. If only the extreme eigenvalues are required, SVD algorithms are better suited.

References

1) Brady, John F., and Georges Bossis. "Stokesian dynamics." *Annual review of fluid mechanics* 20.1 (1988): 111-157.
2) Sierou, Asimina, and John F. Brady. "Accelerated Stokesian dynamics simulations." *Journal of fluid mechanics* 448 (2001): 115-146.
3) Banchio, Adolfo J., and John F. Brady. "Accelerated stokesian dynamics: Brownian motion." *The Journal of chemical physics* 118.22 (2003): 10323-10332.